

Sahtune Fusion Compliance Engine

Architecture Connection Map

A whole-architecture PDF that shows how the FCE system design, seven-gate compliance flow, policy/audit/metadata models, security review, M1-M9 execution architecture, TRL 1-3 sprint plan, laptop PoC validation method, and Claude Desktop handoff connect.

Status	Architecture plus TRL 1-3 build plan. Minimal local PoC code is expected after M1/GATE-A; no production, operational, deployed, classified-processing, or externally installed implementation work is claimed.
Positioning	FCE is a hardened capability inside Sahtune Fusion, not a separate product.
Source	Generated from repo docs/00-16, docs/97-99, and docs/handoff/00-07.
Open blockers	OPEN-01 resolved for M1 Sprint 1. OPEN-02 taxonomy mapping, OPEN-03 edge hardware, OPEN-04 laptop data-source approval, and H1-H14 remain maturity work.

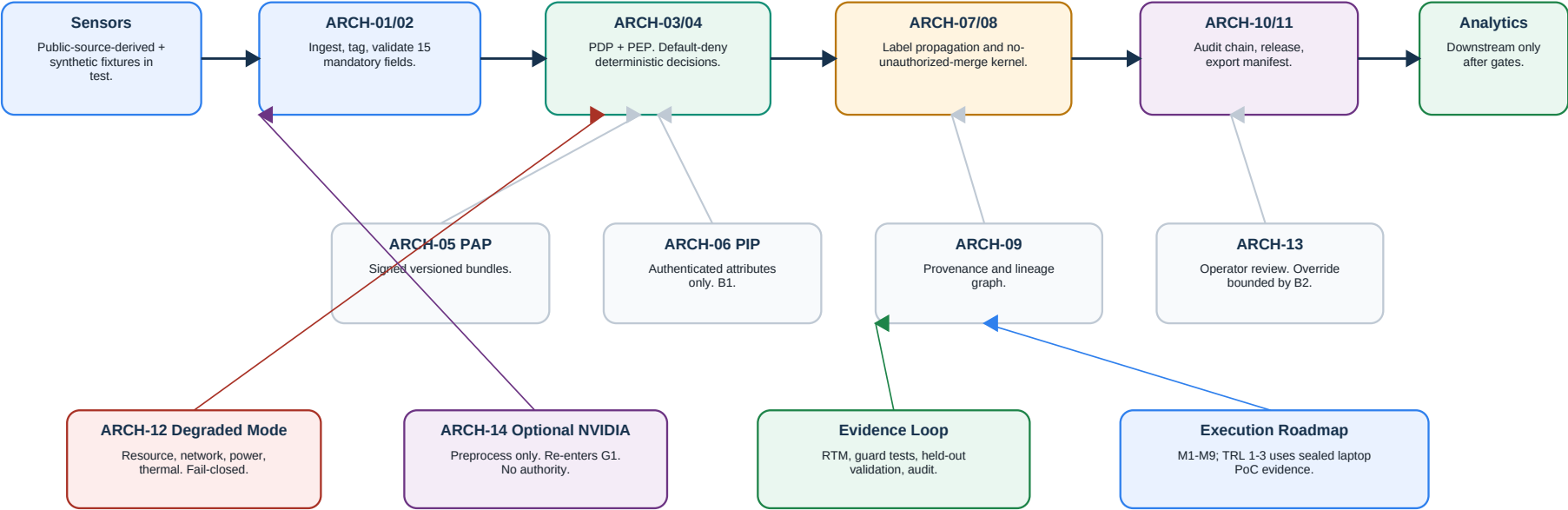
Important claim discipline: no certification, accreditation, ATO, endorsement, classified-processing, or measured-performance claim is made. All performance values remain TARGET until measured on named hardware.

1. Executive View

The FCE inserts deterministic compliance control between sensor data and downstream fusion/analytics.

Whole Architecture Connection Map

Compliance control sits between sensor ingestion and downstream analytics; every path traverses deterministic gates and audit.



Default-deny / fail-closed	Every gate rejects, quarantines, holds, or halts release when inputs are missing, malformed, expired, ambiguous, or unverifiable.
AI advisory only	AI can score, recommend, or detect anomalies, but the final compliance decision cites deterministic policy rules.
No-unauthorized-merge	Fusion output is permitted only if an explicit active-bundle permit covers the combined classification/domain/caveat tuple.
Total auditability	Every decision emits a hash-chained audit record before release/export.

2. Source-of-Truth Architecture Package

The architecture is deliberately block-numbered so each concern can be reviewed and updated without losing traceability.

00-03	Project context, capability decomposition, outcome registry, RTM with 6 Essential and 4 Desired outcomes.
04-08	System architecture, trust boundaries, seven gates, metadata schema, policy decision model, audit schema.
09-13	Synthetic scenarios, threat model, failure modes, TRL roadmap, optional NVIDIA evaluation.
14-16	M1-M9 execution architecture, TRL 1-3 build plan, laptop PoC validation architecture.
97-99	B1-B3 closure, live red-team/security review, design-time gate review.
handoff/00-07	Claude Desktop sprint-loop instructions, prompts, traceability map, and decision register.

System architecture v0	Complete as design draft.
Security-cleared architecture	Not final; H1-H14 remain open and H9 requires later tests.
Proposal-final architecture	Not final until verbatim solicitation text replaces paraphrased anchors.
Execution/sprint architecture	M1-M9 created; TRL 1-3 covers M1-M7 with 14 sprints.
Implementation/code design	Local PoC code starts after M1/GATE-A; production/operational implementation remains deferred.

3. Capability and Requirement Connection

Capabilities translate DND outcome anchors into architecture elements and verification work.

CAP-01 Metadata tagging/validation	ARCH-01, ARCH-02	Supports machine-readable enforcement and ingestion checks.
CAP-02 Deterministic policy enforcement	ARCH-03, ARCH-04	Core default-deny decision path.
CAP-03 Label propagation	ARCH-07	Maintains classification/domain/caveat across derived outputs.
CAP-04 Provenance capture	ARCH-09	Supports lineage for ingested and produced data.
CAP-05 Audit chain	ARCH-10	Documents rules, actions, dispositions, replay.
CAP-06 No-unauthorized-merge	ARCH-08	Blocks unpermitted cross-domain or caveat-unsafe fusion.
CAP-08 Export/release	ARCH-11	Produces integrity manifest and reviewable evidence.
CAP-09 Edge/degraded mode	ARCH-12	Handles SWaP/resource constraints with fail-closed posture.
CAP-10 Operator review/override	ARCH-13	Explainability and bounded override.
CAP-12 Optional NVIDIA evaluation	ARCH-14	Candidate acceleration track only; outside compliance authority.

4. Architecture Elements and Trust Boundaries

Fourteen elements form the system; every service boundary is treated as zero-trust.

Ingest and metadata	ARCH-01, ARCH-02	Accept objects, bind mandatory metadata, validate schema, fail closed at G1/G2.
Policy authority	ARCH-03, ARCH-04, ARCH-05, ARCH-06	Evaluate signed policy bundles using authenticated PIP attributes; apply PDP disposition.
Fusion control	ARCH-07, ARCH-08	High-water-mark labels and no-unauthorized-merge enforcement at G5.
Evidence and release	ARCH-09, ARCH-10, ARCH-11	Provenance graph, audit chain, export/release manifest.
Operations and edge	ARCH-12, ARCH-13	Degraded-mode fail-closed behavior, operator explanation/review/override.
Optional acceleration	ARCH-14	Vision preprocessing candidate only; data-only handoff; re-enters G1.

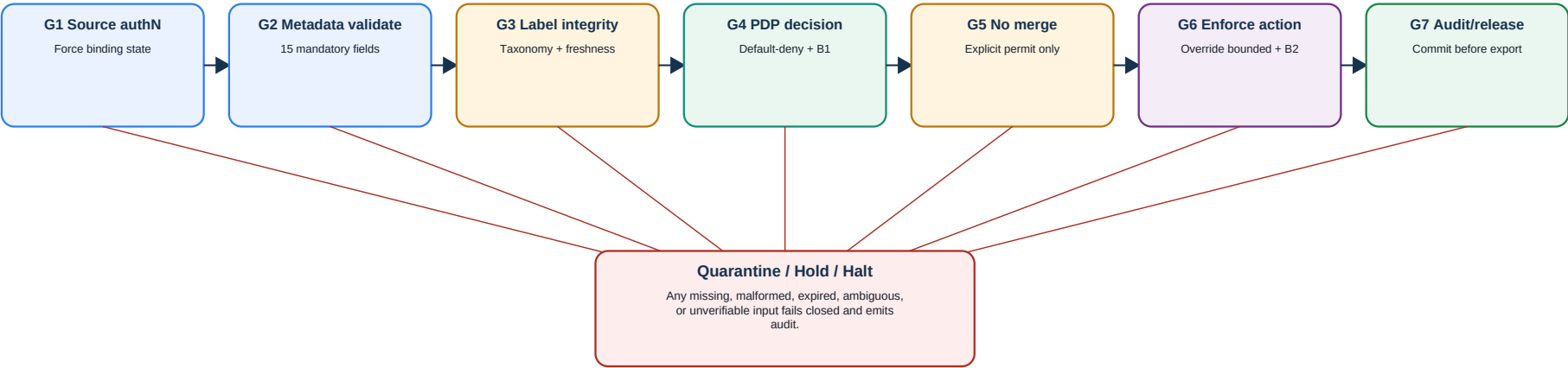
Sensor -> ARCH-01	Source authentication at G1; spoof/missing authN fails closed.
PIP -> PDP	All attributes authenticated and integrity-bound; unverifiable attribute fails closed at G4 with RC-008 (B1).
Operator -> ARCH-13	Authenticated authority; override cannot create permits or relax G5 (B2).
Object binding state -> FCE	policy_binding_state is FCE-authority-set only and forced unvalidated at G1 (B3).
Policy bundle -> PDP/PAP	Signed, versioned bundles; invalid bundles rejected fail-closed.
Audit writer -> store	Append-only, hash-chained, no delete/update path; audit write failure halts release.

5. Seven-Gate Data Flow

Every object traverses all seven gates in order. No path, including GPU acceleration, debug, replay, or admin, bypasses the gates.

Seven-Gate Flow: No Bypass, Fail Closed, Audit Every Decision

Debug, admin, replay, and accelerator paths are not exceptions. Optional acceleration re-enters at G1.



G1	Source authenticated; policy_binding_state reset to unvalidated.	Reject/quarantine unverifiable source.
G2	15 metadata fields present and well-formed.	Reject missing/malformed metadata.
G3	Labels valid; integrity hash and timestamp/freshness valid.	Quarantine invalid, expired, or ambiguous labels.
G4	PDP explicit permit/restriction using authenticated PIP attributes.	Deny/quarantine; RC-008 for unverifiable PIP attribute.
G5	Fusion association has explicit covering permit.	Block merge and segregate inputs.
G6	Disposition applied; override within permitted envelope only.	Reject invalid authority/action.
G7	Audit record appended; export manifest generated if releasing.	Halt release if audit write fails.

6. Metadata, Policy, and Audit Work Together

The FCE decision loop depends on a stable data contract, a deterministic policy model, and replayable audit evidence.

Metadata schema	15 mandatory fields; project taxonomy; source values cannot set validated policy state.	FCE-REQ-MET-010, PRV-001/002, POL-011. Unit/integration tests later.
Policy model	PDP/PEP/PAP/PIP; default deny; 11 actions; RC-001..RC-008; deny-overrides; version-pinned bundles.	FCE-REQ-POL-001/011/012/020, KRN-010, OPS-002. Property/red-team tests later.
Audit model	18 mandatory fields; 9 event classes; append-only hash chain; deterministic replay; export manifest.	FCE-REQ-AUD-001/002/003, EXP-001. Replay and chain tests later.
Fusion kernel	Explicit-permit merge check; high-water-mark propagation; parent linkage in provenance.	FCE-REQ-KRN-010, POL-012. No-merge red-team tests later.

AI advisory only	The system can record confidence or anomaly signals, but it must proceed deterministically without AI and cite rule IDs.
High-water mark	Merged/derived objects inherit the most restrictive parent labels unless an authorized downgrade proof applies.
Audit before release	If G7 cannot durably write the decision record, release/export halts.

7. Security Review and Risk State

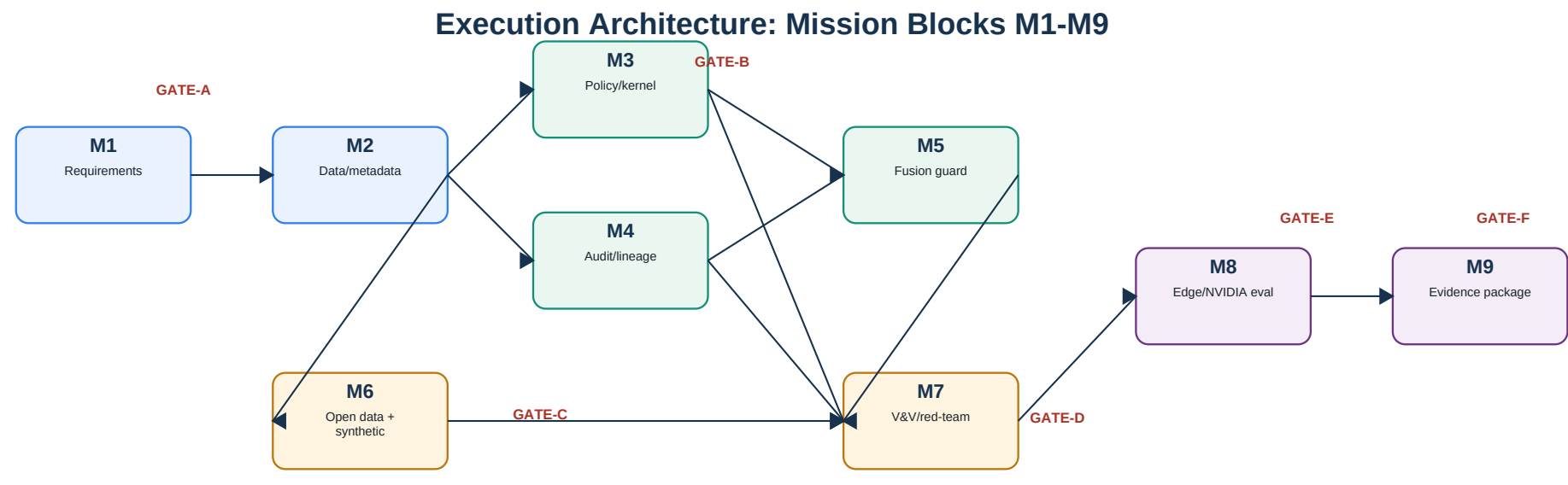
The live red-team and security-assurance review approved the design with conditions, not as final security clearance.

B1	Closed in text; test later	PIP attributes must be authenticated/integrity-bound; unverifiable fails closed at G4 with RC-008.
B2	Closed in text; test later	Override cannot relax no-unauthorized-merge or cross-domain block.
B3	Closed in text; test later	policy_binding_state is FCE-authority-set only and reset at G1.
H1-H14	Open	High-priority maturity items before TRL 4-5 exit.
H9	Open test proof	Demonstrate no-bypass, no-unauthorized-merge, and B1-B3 by property/red-team tests.
Claim audit	Clean at v0	No prohibited claim found in the design package.

Fusion authority	ARCH-08 sole fusion authority; detect empty-parentage evasion.
Crypto and audit hardening	Root-of-trust, key management, audit anchoring, total-order audit writer.
Time and update safety	Trusted time source, anti-replay, emergency revocation, anti-rollback.
Edge protection	At-rest protection, secure boot, physical tamper posture.
Governance	Two-person policy publication, release destination authentication, audit export authorization.

8. Execution Architecture: M1-M9

The M-blocks convert the system architecture into a staged plan with review gates.



TRL 1-3 covers M1-M7 only; M8 and M9 are planned but later execution depends on approval, named hardware, and evidence gates.

GATE-A	Requirements locked with verbatim solicitation and full RTM coverage.	After M1
GATE-B	Schema, policy, and audit design coherent.	After M2, M3, M4
GATE-C	Public-source manifest, trim report, fixture seal, and synthetic red-team variants ready.	After M6
GATE-D	Guard rejection tests, held-out laptop validation, no-bypass/no-merge evidence ready.	After M5, M7
GATE-E	Edge benchmark and NVIDIA evaluation plan ready.	After M8
GATE-F	Proposal/evidence package ready with clean claim screen.	After M9

9. TRL 1-3 Sprint Plan

The first execution band is design plus minimal local proof-of-concept: M1-M7, two sprints per block, 14 total sprints.

M1 Requirements	Finalized RTM rows draft	Coverage report and GATE-A decision	GATE-A
M2 Data/metadata	Schema and provenance design	Schema-validation PoC plus tests	GATE-B partial
M3 Policy/kernel	Policy model and rule examples	Deterministic policy-evaluator PoC plus tests	GATE-B partial
M4 Audit/lineage	Audit/provenance schema design	JSONL audit/provenance PoC plus replay check	GATE-B partial
M5 Fusion guard	Fusion-kernel interface spec	No-unauthorized-merge PoC plus blocked-merge evidence	GATE-D partial
M6 Open data + synthetic	Scenario/source/trim plan	Source manifest, trim report, calibration/held-out fixtures, fixture seal	GATE-C
M7 V&V/red-team	V&V; matrix and red-team plan	Guard tests plus held-out laptop validation report	GATE-D partial

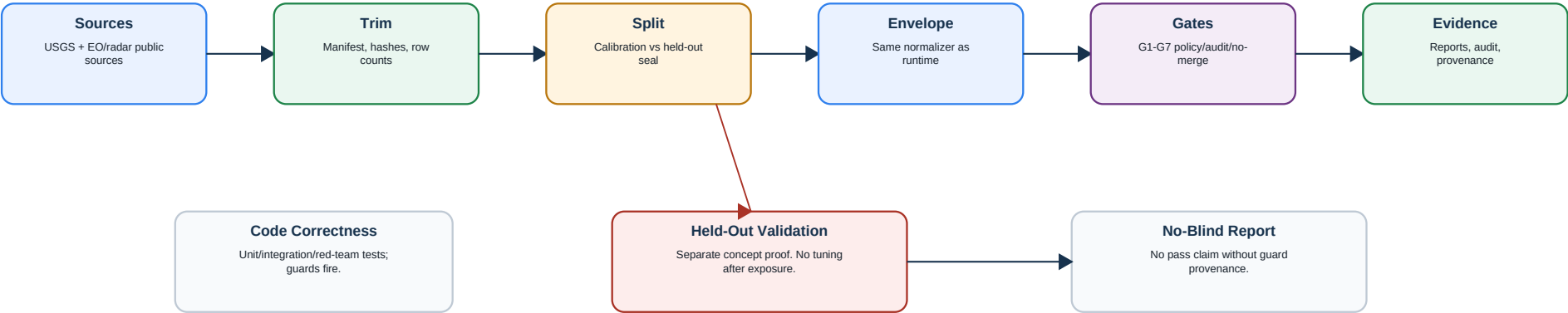
Production or operational implementation	TRL 1-3 produces local PoC evidence only; deployable implementation needs later approval.
Sensor adapters and live/private/controlled/classified data	Current work uses approved public-source-derived fixtures plus synthetic variants only.
NVIDIA component installation	Paper evaluation only until approved; optional track outside compliance path.
Measured performance claims	Require named hardware, provenance, and benchmark execution.
Production cryptography/accreditation	Current signatures/hashes are design placeholders and support artifacts only.

10. Laptop PoC Validation Architecture

This is the pre-lab proof method: source manifest, trim, sealed fixtures, guard tests, and held-out validation.

Laptop PoC Validation: Make Invalid Proof Hard

Green unit tests are code correctness, not concept validation. Held-out evidence is reported separately.



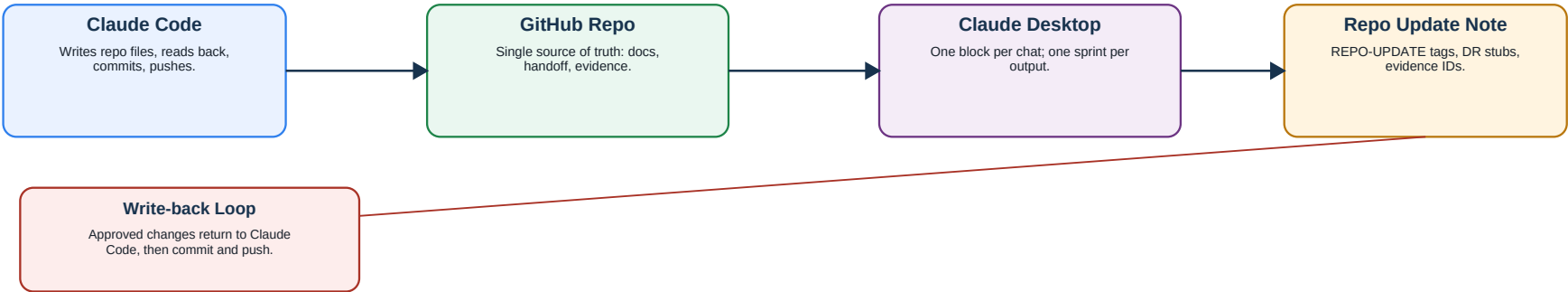
Code correctness	Schema, policy, audit, merge guard, and invalid-proof guards behave as specified.	Real-world mission suitability.
Held-out laptop validation	Integrated FCE flow works on sealed public-source-derived + synthetic fixtures.	Lab, edge, field, classified, or operational performance.
Lab/edge testing	Later TRL 4-5 feasibility on named hardware.	Certification, ATO, or operational authority.

Source/trim guards	Missing URL, hash, licence note, query params, row counts, or split seed blocks fixture proof.
Contract guards	Unsupported envelope/policy/evidence versions are rejected instead of silently accepted.
Compliance guards	PIP spoofing, source pre-validation, no-bypass, no-merge, override abuse, and audit-write failure all reject.
Reporting guards	Diagnostic mode cannot write evidence; no report is emitted without guard provenance.

11. Claude Desktop Review Loop

Desktop can now review each sprint without becoming the source of truth.

Claude Desktop Continuation Without Losing Traceability



Desktop is for review/planning. It does not own the architecture unless Claude Code writes the change back into the repo.

One mission block per chat	Keeps context narrow and reviewable.
One sprint per output	Preserves the M1-M7 / S1-S2 structure.
Cite repo paths and requirement IDs	Prevents untraceable architecture drift.
Tag repo changes as REPO-UPDATE	Claude Code can apply, read back, commit, and push.
PoC code only after M1/GATE-A	Desktop drafts changes; Claude Code writes files, runs local tests, commits, and pushes.
Stop if source-of-truth data is missing	Do not invent repo state, requirement IDs, outcome text, or measured results.

12. Decision Register and Next Practical Action

These are the decisions that must be resolved as the architecture moves into sprint review.

OPEN-01: verbatim DND solicitation text	Resolved for M1 Sprint 1; outcome anchors are verified in docs/02 and docs/03.	Closed for M1 S1
OPEN-02: project taxonomy mapping	Defines how project labels map to named handling target without using real markings.	M2 / M3
OPEN-03: edge hardware SKU	Keeps benchmarks as TARGET until named hardware and tests exist.	M8 / TRL 4-5
OPEN-04: laptop PoC sources	Select public source families, trim limits, held-out split, and guard list before fixture generation.	M6 / M7
H1-H14 triage	Assigns owners and maturity band for high-priority security conditions.	Before TRL 4-5 exit
TRL 1-3 PoC boundary	Authorizes local PoC code plus sealed public-source-derived/synthetic evidence after M1/GATE-A.	M2-M7
External evaluation approval	Required before NVIDIA or other external tools are installed/tested.	M8

Recommended immediate path

Continue with M1 Sprint 2 and GATE-A readiness. After GATE-A, run M2-M5 code-correctness work, M6 source/trim/fixture sealing, and M7 held-out laptop validation before lab testing. Desktop drafts REPO-UPDATE notes; Claude Code writes, runs, commits, and pushes.

13. Appendix: Source References

The PDF summarizes the architecture; these files remain the authoritative source.

Concept and requirements	docs/01, docs/02, docs/03
System architecture	docs/04, docs/05
Data/policy/audit	docs/06, docs/07, docs/08
Scenarios/security/failure modes	docs/09, docs/10, docs/11
TRL and optional acceleration	docs/12, docs/13
Execution and laptop PoC validation	docs/14, docs/15, docs/16
Review and closure	docs/97, docs/98, docs/99
Desktop handoff	docs/handoff/00 through docs/handoff/07

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